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20 May 1988
SUPERSEDING
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27 April 1979

MILITARY SPECIFICATION

DOLLY, TRAILER CONVERTORS, M197A1, M198A1 AND M354

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M197A1, M198A1 and M354 tactical dollies for use in converting various semitrailers to full trailers.

1.2 Classification. Dollies, also referred to herein as "vehicles", shall be classified by model as follows (see 6.2 and 6.5):

Model	Description
M197A1	Dolly, Trailer Convertor, 6 Ton, 2 Wheel
M198A1	Dolly, Trailer Convertor, 8 Ton, 2 Wheel
M354	Dolly, Trailer Convertor, 15 Ton, 4 Wheel. W/E

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
FEDERAL

VV-B-680	- Brake Fluids, Automotive.
VV-L-800	- Lubricating Oil, General Purpose, Preservative, (Water-Displacing, Low Temperature).
ZZ-I-550	- Inner Tube, Pneumatic Tire.

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-3150	- Lubricating Oil, Preservative, Medium.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-T-12459	- Tire Pneumatic: For Military Ground Vehicles.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather Operational and Preservative.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-C-53039	- Coating, Aliphatic Urethane, Single Component, Chemical Agent Resistant.
MIL-V-62038	- Vehicles Wheeled: Preparation for Shipment and Storage of.

STANDARDS
FEDERAL

FED-STD-595	- Color.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Standard (FMVSS)

FMVSS NO. 105	- Hydraulic Brake System.

DRAWINGS
ARMY

8736120	- Dolly, Trailer Convertor, 6 Ton, 2 Wheel, M197A1.
8736124	- Dolly, Trailer Convertor, 8 Ton, 2 Wheel, M198A1.
8736308	- Dolly, Trailer Convertor, 15 Ton, 4 Wheel, M354.
12355846	- Treatment and Paint Specifications for Tactical Trailers.

(Copies of specifications, standards, drawings, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersed applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish dollies, of each model to be delivered under the contract which shall be subjected to first article inspection. First article inspection samples, properly marked with identifying information, shall be representative of the unit to be furnished to the Government. All subsequent dollies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes (see 4.3).

3.1.1 Reliability. Dollies with rated payload, when operated in accordance with table I, shall possess not less than 60 percent (%) probability of completing not less than 6000 miles of operation without replacement or overhaul of the principal components and/or subassemblies, that is, suspension, including springs, axles and wheels, major brake system components (less brake shoes and common hardware), landing device, lunette, drawbar and frame (see 4.8.18). The criteria for replacement or overhaul shall be as prescribed by the maintenance manual. The 6000 miles of operation shall be apportioned as follows:

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a.	Hard-surfaced roads	40%
b.	Munson improved gravel roads	25%
c.	Level cross-country	24%
d.	Hilly cross-country	10%
e.	Belgian block	1%

TABLE I. Vehicle loads and speeds.

Model	Rated Payload 1/ (pounds)	Total 2/ weight (pounds)	Towing vehicle	Semitrailer class	Maximum speeds (mph)	
					Highway	Cross-country
M197A1	12,000	14,880	2 1/2 ton	6 ton	50	10
M198A1	16,000	12,505	5 ton	8 ton	50	10
M354	30,000	36,100	10 ton	15 ton	35	10

1/ Payloads indicated are for highway and cross-country operations. If the dolly and the empty semitrailer reach the total combined weight requirement no semitrailer payload is required.
2/ Combined weight of dolly and payload.

3.1.2 Maintainability. The total scheduled maintenance, excluding driver/crew checks and services, shall be between 4 and 8 manhours. The total unscheduled maintenance shall be between 0 and 12 manhours during the first 6000 miles of operation (see 4.8.19). This equates to a Maintenance Ratio (MR) of 0.013 to 0.067 at 20 miles per hour (mph) and a MR of 0.02 to 0.10 at 30 mph. The scheduled maintenance intervals shall be 3 months or 3000 miles of operation, whichever comes first. Maximum time to repair (only to include diagnosis, repair and verification) shall be as follows, using personnel normally employed:

a. Organization maintenance shall not exceed 3 manhours 95% of the time.
b. Direct support maintenance shall not exceed 10 manhours 95% of the time.
c. General support maintenance - not required.

3.2 Materials. Materials shall be as specified herein and in referenced drawings, specifications, and standards (see 4.8.1).

3.2.1 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings (see 4.8.1).

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3.2.2 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor's inspection records shall list all QPL items by number and date of the QPL, name of the supplier and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of the supplier(s) (see 4.1.4 and 4.8.1).

3.2.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.3 Construction. The vehicle shall be constructed in accordance with the applicable drawings specified in table II, and as supplemented herein. If more than one unit of a particular component is used in or on the vehicle, the components used shall be identical in make, material, and quality. The riveting, welding practices, and quality shall be the same for each vehicle (see 4.8.1 and 4.8.2).

TABLE II. Drawings.

Model	Drawing
M197A1	8736120
M198A1	8736124
M354	8736308

3.3.1 Seals. When fording or operating in mud, sand or snow, seals shall prevent entrance of foreign matter into the bearings. Bearing seals shall restrict the leaking of lubricants from bearing (see 4.8.3).

3.3.2 Electrical circuits. Electrical circuits shall maintain continuity from end to end, without evidence of internal or external shorts, during all vehicle operating conditions (see 4.8.2 and 4.8.4).

3.3.3 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction under all vehicle operating conditions (see 4.8.2 and 4.8.5).

3.3.4 Adjustment mechanisms. All adjustment mechanisms shall function properly, and maintain adjustment within required settings during all vehicle operating conditions (see 4.8.2 and 4.8.6).

3.3.5 Welding. Welding shall meet the quality requirements specified on the applicable drawing (see 4.8.2 and 4.8.7).

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3.3.5.1 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the procuring activity (see 6.2).

3.3.6 Tires and tubes.

3.3.6.1 Tires. Tires shall be class CC, style A, of MIL-T-12459 (see 4.8.1 and 4.8.2).

3.3.6.2 Inner tubes. Inner tubes shall conform to group II of ZZ-I-550, of proper size for the tire specified (see 4.8.1 and 4.8.2).

3.4 Performance. The vehicle shall meet the performance requirements of this specification when fully equipped, loaded in accordance with table I, serviced with products specified in table III, and towed by appropriate military towing vehicle.

TABLE III. Service products.

Product use	Ambient air temperature	
	-65°F to 0°F	-10°F to +132°F
Oil:		
For hydraulic brakes	MIL-B-46176	VV-B-680
For general purpose lubrication	VV-L-800	MIL-L-3150
For shock absorbers	MIL-H-5606	MIL-H-5606
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis lubrication, including wheel bearings	MIL-G-10924	MIL-G-10924

3.4.1 Environmental. The vehicle shall operate as specified herein, under extreme conditions of weather, in ambient air temperatures ranging from minus 50 degrees Fahrenheit (°F) to plus (+) 125°F (see 4.8.8).

3.4.2 Service products. Servicing and lubrication shall be performed using standard service products specified in table III (see 4.8.1).

3.4.3 Highway and cross-country operations.

3.4.3.1 Highway operation. When loaded and towed at speeds specified in table I, in a straight line, on a relatively level, smooth paved surface, the path of the towed vehicle shall not deviate more than 3 inches to either side of the path of the towing vehicle (see 4.8.9).

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3.4.3.2 Cross-country operation. The vehicle shall be towable at specified load and speed (see table I) over unimproved roads, or gently rolling terrain. When so operated in a straight line, the vehicle shall track the towing vehicle without damage to itself or the towing vehicle. Stability shall be maintained (see 4.8.10).

3.4.4 Gradeability. When towed over a dry, hard-surfaced road, free of loose material, each vehicle shall exhibit stability and gradeability specified below (see 4.8.11).

3.4.4.1 Longitudinal inclines. Each vehicle shall follow the towing vehicle, without weaving to an extent which adversely affects the controllability of the vehicle combination, when ascending or descending longitudinal inclines having a 15% grade (4.8.11).

3.4.4.2 Side slopes. Each vehicle shall follow the towing vehicle without slipping or upsetting when operating, either left or right, on 20% side slopes (see 4.8.12).

3.4.5 Turning ability. When coupled to the towing vehicle, operating in its minimum turning circle, each vehicle shall follow without cramping or side-slipping and without damage to or interference between the towed vehicle and the towing vehicle (see 4.8.13).

3.4.6 Braking ability.

3.4.6.1 Service brakes. The combined service brakes of the vehicle and towing vehicle shall control, decelerate, and stop the vehicle combination within 30 feet from a speed of 20 mph on dry, hard, level, smooth surfaces roads. Hydraulic brakes shall conform to FMVSS No. 105 (see 4.8.2 and 4.8.14).

3.4.6.2 Automatic actuation. The automatic breakaway device shall apply and maintain application of the vehicle brakes for not less than 15 minutes, upon breakaway from the towing vehicle (see 4.8.2 and 4.8.15).

3.4.7 Fording. Each vehicle shall ford hard-bottomed, salt or fresh water crossings of depths great enough to completely submerge the chassis to top of fifth wheel and shall withstand such submergence for periods up to 15 minutes duration without damage to the vehicle or its components (see 4.8.16).

3.4.8 Landing leg. Unless otherwise specified (see 6.2), the vehicle shall have a landing leg. The landing leg shall exhibit capacity and adjustment to raise, support and hold its portion of the fully loaded vehicle and shall withstand, without damage, the strains imposed when coupling or uncoupling the towing vehicle. The landing leg shall extend and retract; and when retracted, shall provide ground clearance for cross-country operation (see 4.8.2 and 4.8.17).

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3.5 Painting. Pretreatment, cleaning, priming and painting shall be in accordance with MIL-STD-193 and Drawing Number 12355846.

3.6 Marking. Marking of the vehicle shall be in accordance with the applicable drawing and MIL-STD-642 (see 4.8.2).

3.6.1 Identification plates. Identification plates shall meet the requirements of MIL-P-514 (see 4.8.2).

3.6.2 Components. All parts, subassemblies and assemblies requiring identification shall be identified in accordance with MIL-STD-130 (see 4.8.2).

3.7 Safety. All exposed parts which are energized electrically shall be located, insulated, fully enclosed, or guarded so as to prevent hazards to operating or maintenance personnel, or interference with equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions (see 4.8.20).

3.8 Servicing. The vehicle shall be adjusted for immediate operation and shall be serviced for climatic conditions at the delivery point. Servicing shall be compatible with processing for shipment as specified by the procuring activity (see 4.8.2, 4.8.21 and 5.1).

3.9 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit or reduce the capability of the vehicle in the performance of its intended use. In addition, the vehicle shall be free of defects which affect its appearance (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until the conformance of the product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a QPL, the contractor shall make available to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity.

4.1.5 Quality assurance provision. In the conduct of inspection, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Provision (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.6 Certification. Where certification (see 4.8.1) is required to verify material or component conformance to a specification, the contractor shall maintain such certification along with documented test results and performance and analytical data, as applicable.

4.2 Classification of inspection:

a. First article inspection (see 4.3).
   1. First preproduction vehicle inspection (FPVI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

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b. Quality conformance inspections (QCI) (see 4.4).
c. Control tests (see 4.5).
d. Comparison tests (see 4.6).

4.3 First article inspection. First article inspections shall be performed on the first production and on two additional vehicles randomly selected from the first month's production, or from the first 20 produced, which shall be designated as initial production vehicles. Approval of the first article samples shall not relieve the contractor of the obligation to supply vehicles that are fully representative of those inspected as a first article sample. Any change or deviation of production vehicles from the first article sample shall be subject to approval of the contracting officer.

4.3.1 FPVI.

4.3.1.1 In-process inspection. The first production vehicle shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. The vehicle's suspension and electrical components shall be available for inspection. Inspection shall be made prior to application of primer and paint. Processing and welding procedures, quality system and inspection records, calibration procedures, radiographic procedures and welder certification shall be reviewed and evaluated.

4.3.1.2 Completed vehicle inspections.

4.3.1.2.1 Contractor inspection. The contractor shall subject the first production vehicle of a specific model to the 50-mile road test specified in 4.5.1 and to FPVI as specified in table IV.

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element will conduct the FPVI specified in table IV to the extent that acceptability of the vehicle is verified.

4.3.1.2.3 Provisional inspection. The Government procuring activity will perform the FPVI specified in table IV to the degree required to justify approval of the vehicle.

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TABLE IV. Classification of inspections.

Title	Requirements	Inspection	First article		1/ QCI	1/ Control	2/ Comparison
			1/ FPVI	2/ IPT			
Materials and construction	3.2 thru 3.3, 3.3.6 and 3.4.2	4.8.1	X				X
Defects	3.3, 3.3.2 thru 3.3.6, 3.4.6, 3.4.8, 3.5 thru 3.6.2, 3.8 and 3.9	4.8.2	X	X	X		X
Reliability	3.1.1	4.8.18		X			
Maintainability	3.1.2	4.8.19		X			
Seals	3.3.1	4.8.3	3/	3/			3/
Electrical circuits	3.3.2	4.8.4	3/	3/	3/	3/	3/
Controls	3.3.3	4.8.5	3/	3/	3/	3/	3/
Adjustment mechanism	3.3.4	4.8.6	3/	3/	3/	3/	3/
Welding	3.3.5	4.8.7	3/	3/	3/		
Environmental	3.4.1	4.8.8	X	X			
Highway operation	3.4.3.1	4.8.9	X	X	X	X	X
Cross-country operation	3.4.3.2	4.8.10	X	X			X
Longitudinal inclines	3.4.4.1	4.8.11	X	X			X
Side slopes	3.4.4.2	4.8.12	X	X			X
Turning ability	3.4.5	4.8.13	X	X		X	X
Service brakes	3.4.6.1	4.8.14	X	X	X	X	X
Automatic actuation	3.4.6.2	4.8.15	X	X		X	X
Fording	3.4.7	4.8.16	X	X			X
Landing legs	3.4.8	4.8.17	X	X	X	X	X
Safety	3.7	4.8.20	3/	3/			3/
Servicing actuation	3.8	4.8.21	3/	3/	3/	3/	3/

1/ At place of manufacture
2/ At Government proving ground
3/ These examinations shall be part of the applicable road test.

4.3.1.2.5 Vehicle disposition. After completion of inspection, the first production vehicle shall remain at the manufacturing facility, as a production sample, and be the last vehicle of its type shipped on the contract. The vehicle may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the vehicle during this period in accordance with applicable documents for care and

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preservation while in storage. All configuration changes taking place after the first production inspection, shall be made to the first production vehicle so that the vehicle will be representative of the current configuration throughout the life of the contract, no configuration changes may be implemented on production vehicles after Government conditional acceptance of the first production vehicle, of the applicable vehicle model, without written authorization from the Government.

4.3.2 IPT.

4.3.2.1 Contractor inspection. The contractor shall inspect the IPT vehicles in accordance with 4.4.

4.3.2.2 Government inspection. The Government will perform the IPT specified in table IV and a 6000 mile test in accordance with table V, at a site approved by the Government. The contractor shall furnish repair parts, as required to support the tests at no cost to the Government.

TABLE V. 6000 mile test.

Course	Mileage and speeds 1/
Hard-surfaced, paved	2400 miles at varying speeds up to maximum highway speed with rated payload
Hilly cross-country (Churchville "C")	600 miles at varying speeds up to maximum cross-country speed with rated payload
Level cross-country	1440 miles at varying speeds up to maximum cross-country speed with rated payload
Munson improved gravel roads	1500 miles at varying speeds up to maximum cross-country speed with rated payload
Belgian block	60 miles at varying speeds up to maximum cross-country speed (40 miles shall be with rated payload and 20 miles shall be with no payload)

1/ See table I for speeds and payloads.

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4.3.3 Deficiencies. Unresolved failures or deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the IPT shall be evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all vehicles shall be corrected by the contractor.

4.4 QCI. QCI shall be performed in the following sequence and utilizing an approved Final Inspection Record (FIR).

a. Examination for the defects specified in table IV and classified in table VI.
b. A 5 mile road test in accordance with 4.4.4.
c. QCI specified in table IV.

4.4.1 One hundred percent. Each vehicle shall be subjected to the QCI specified in 4.4.

4.4.2 Waiver of 100% QCI. When authorized by the Government contracting authority, 100% QCI (see 4.4.1) shall be waived. Thereafter, samples shall be selected in accordance with 4.4.2.1 and shall be subjected to the QCI specified in 4.4. For examination purposes samples shall be inspected to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	25/100 units
Minor	150/100 units

4.4.2.1 Sampling.

4.4.2.1.1 Lot formation. An inspection lot shall consist of all the vehicles of one model, from one day's production, submitted at one time for acceptance.

4.4.2.1.2 Sampling for inspection. When sampling is authorized for QCI (see 4.4.2) samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall examine at least 20 consecutively produced vehicles. The examination shall establish that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQL.

4.4.3 Defects. The following rules shall be applied to the counting of defects (see 6.4).

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4.4.3.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots, or 10 lots or more within a 30-day period, will be added to the minor defects classification with no increase in AQLs.

4.4.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

TABLE VI. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 25 defects per 100 units	
101	Safety chains, intervehicular cable and hose - Improper lengths, damaged, malfunctioning, missing (see 3.3).	Visual, functional and SIE 1/
102	Suspension system components, springs and shock absorbers - Damaged, leaking, noisy (see 3.3).	Visual and functional
103	Electrical systems - Malfunctioning, inoperative, damaged (see 3.3.2).	Visual and functional
104	Controls, adjustment mechanisms - Malfunctioning, inoperative, damaged (see 3.3.3 and 3.3.4).	Visual and functional
105	Frame - Structural welding and riveting defects (see 3.3 and 3.3.5).	Visual
106	Wheels, tires, tubes - Not as specified, tire or tube damage (see 3.3.6).	Visual and SIE
107	Axle - Improper assembly or installation welding defects (see 3.3 and 3.3.5).	Visual and functional
108	Drawbar assembly - Raising or lowering action improper, welding defects (see 3.3 and 3.3.5).	Visual and functional
109	Fifth wheel - Malfunctioning, improper assembly or installation, welding defects (see 3.3 and 3.3.5).	Visual and functional
110	Landing leg - Improper extension retraction or stowage action, and locking action (raised and lowered) (see 3.4.8).	Visual and functional
111	Lunette eye - Casting defect, improper size (see 3.3).	Visual and SIE

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TABLE VI. Classification of defects - Continued.

Category	Defect	Method of examination
112	Brake system - Malfunctioning, component damage, nonconformance (see 3.4.6).	Visual and functional
113	Workmanship affecting performance (see 3.9).	Visual
Minor	AQL 150 defects per 100 units	
201	Reflectors - Missing, defective, Improper installation (see 3.3).	Visual
202	Electrical system components - Improper assembly or installation (see 3.3.2).	Visual
203	Wiring and tubing - Improperly coded protected, installed or assembled (see 3.3.2 thru 3.3.4).	Visual
204	Wheels, tires and tubes - Improper mounting, pressure or stud torque (see 3.3.6).	Visual or tactile
205	Drawbar assembly - Improper assembly or installation (see 3.3).	Visual
206	Landing leg - Improper assembly or installation (see 3.3 and 3.4.8).	Visual
207	Lunette eye - Improper assembly or installation (see 3.3).	Visual
208	Suspension system components; springs and shock absorbers - Improper assembly or installation (see 3.3).	Visual
209	Tie downs, lashing hooks and lifting devices Missing, improperly located, welding defects (see 3.3).	Visual
210	Brake components - Improper assembly or installation, clearance inadequate (see 3.4.6).	Visual
211	Paint - Improper application or color (see 3.5).	Visual
212	Marking - Missing, incomplete, improper size (see 3.6).	Visual
213	Identification plates and identification markings - Missing, incomplete, improper size, improper installation (see 3.6.1 and 3.6.2).	Visual
214	Lube fittings - Missing, improper installation, defective (see 3.3).	Visual
215	Lubrication - Improper application (see 3.8).	Visual
216	Protective coating - Application or coverage improper (see 3.3).	Visual
217	Record forms - Missing information, improperly processed (see 3.3).	Visual
218	Tools - Missing, improperly processed (see 3.3).	Visual
219	Workmanship affecting appearance (see 3.9).	Visual

1/ SIE = Standard Inspection Equipment.

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4.4.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.4.4 Road test. The vehicle, without payload, shall be operated over a 5-mile course of relatively, hard-surfaced roads.

4.5 Control tests. The Government shall select, at random, 1 production vehicle per month as a minimum, or 1 out of each 100 vehicles produced per month, but not more than 2 in any 30 day periods. The contractor shall conduct the 50-mile road test specified in 4.5.1 on each vehicle selected. Subsequently, the vehicle shall be subjected to all control tests specified in table IV.

4.5.1 Fifty-mile test. Each vehicle selected shall be loaded with a full or simulated load and shall be operated for a distance of 50 miles. The test course shall be relatively smooth, approximately level, hard-surfaced roads.

4.6 Comparison test. The Government may randomly select vehicles at anytime during the production contract period and subject these vehicles to the comparison tests specified in table V as well as a 6000 mile test in accordance with table IV. Tests will be conducted by the Government at a site it will select. Vehicles selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any vehicle to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3, 3.3.6 and 3.4.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.3.2 through 3.3.6, 3.4.6, 3.4.8, 3.5 through 3.6.2, 3.8 and 3.9 shall be determined by examination for the defects listed in table VI. Examination shall be visual, tactile, functional or by measurement with standard inspection equipment.

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4.8.3 Seals. To determine conformance to 3.3.1, after the 6000 mile test and fording operation, all bearings shall be checked for dirt contamination and lubricant leakage.

4.8.4 Electrical circuit. To determine conformance to 3.3.2, current and voltage output levels and continuity of the electrical system shall be measured.

4.8.5 Controls. To determine conformance to 3.3.3, all controls shall be operated and examined for functional requirements.

4.8.6 Adjustment mechanisms. To determine conformance to 3.3.4, all adjustment mechanisms shall be examined for required adjustment and function.

4.8.7 Welding, repairs and burning operations. To determine conformance to 3.3.5 and 3.3.5.1, all welding, welding repairs and burning operations shall be examined to determine that slag has been removed and approved procedures have been followed.

4.8.8 Environmental. To determine conformance to 3.4.1, the vehicle, properly serviced and equipped, shall be operated at the tempertures specified. The vehicle shall evidence no damage as a result of such operation.

4.8.9 Highway operation. To determine conformance to 3.4.3.1, the vehicle shall be coupled to the towing vehicle. The combination shall be operated on paved highway and shall meet the performance specified.

4.8.10 Cross-country operation. To determine conformance to 3.4.3.2, the vehicle shall be operated over cross-country terrain, as specified, without damage to the vehicle or towing vehicle.

4.8.11 Longitudinal inclines. To determine conformance to 3.4.4.1, the vehicle shall be towed up and down longitudinal inclines at cross-country speed and shall meet specified requirements.

4.8.12 Side slopes. To determine conformance to 3.4.4.2, the vehicle shall be towed on side slopes at cross-country speed and shall meet specified requirements.

4.8.13 Turning ability. To determine conformance to 3.4.5, the towing vehicle shall be coupled to the vehicle, driven to the towing vehicle's minimum turning circle, to the right and to the left, without interference between the towing and the towed vehicles.

4.8.14 Service brakes. To determine conformance to 3.4.6.1, the vehicle combination shall be operated as specified and observed for stopping distance. Stopping distance shall not exceed that specified. During QCI, the vehicle braking system shall be operated to the extent necessary to insure proper operation and adjustment of service brakes on all wheels.

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4.8.15 Automatic actuation brake. To determine conformance to 3.4.6.2, the vehicle and towing vehicle shall be placed on 15% grade and the automatic break-away device actuated. The brakes shall hold the vehicle stationary for time specified, when uncoupled from towing vehicle.

4.8.16 Fording. To determine conformance to 3.4.7, the vehicle shall be towed through water crossing at the depth and for the time period specified.

4.8.17 Landing leg. To determine conformance to 3.4.8, the vehicle shall be coupled and uncoupled from the towing vehicle and the landing leg operated. The landing leg shall perform as specified.

4.8.18 Reliability. To determine conformance to 3.1.1, reliability requirements shall be verified at a 50% confidence level during the 6000 mile IPT test specified in 4.3.2.2.

4.8.19 Maintainability. To determine conformance to 3.1.2, the maintainability requirements shall be verified during the 6000 mile IPT test specified in 4.3.2.2.

4.8.20 Safety. To determine conformance to 3.7, the vehicle shall be examined to assure that hazards to operating and maintenance personnel have been eliminated or controlled.

4.8.21 Servicing. To determine conformance to 3.8, the vehicle shall be examined for proper servicing and adjustment of specified systems.

5. PREPARATION FOR DELIVERY.

5.1 Preservation/processing. Preservation/processing for shipment and storage shall be in accordance with MIL-V-62038 and the applicable Equipment Preservation Data Sheet for the level(s) specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Dollies covered by this specification are intended for use in converting semitrailers to full trailers for transporting military supplies and equipment, and for mounting special equipment, when required.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Model of required item (see 1.2).
c. If first article inspection is not required (see 3.1).
d. Whether welding repair is as authorized, and extent allowed (see 3.3.5.1).
e. If a landing leg is not required (see 3.4.8).
f. If responsibility for inspection shall be other than as specified (see 4.1).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).

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h. Selection of applicable level and equipment preservation requirements.

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Deficiency sheet. The Government inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any departure from good workmanship. The Government inspector will assure all deficiencies encountered during the inspection are enumerated on the deficiency sheet for the vehicle. The defects noted on the deficiency sheet should contain sufficient description to enable the Government inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects (see table VI and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.5 Vehicle characteristic data. Vehicle characteristic data for the dollies covered by this specification is provided in the following MS sheet form standards.

Model	MS standard
M197A1	MS53025
M198A1	MS53026
M354	MS500081

6.6 Subject term (key word) listing.

Dolly
Trailer convertors
Trailer, dolly

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6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
Navy - MC

(Project 2330-0983)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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FSC 2330

DOCUMENT IDENTIFIER AND TITLE MIL-D-45322F; Dolly, Trailer Convertors, M197A1, M198A1, And M354

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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